

Milestone 1: Preprocessing Analysis Results

Abdullah Aldahleh 20220385

Tala Beirouti 20230450

This report analyzes the dataset before preprocessing to understand its structure and content. The dataset includes different document types and covers multiple technical topics. The data was collected from different sources and includes documents of different lengths. The analysis was performed using computational methods to extract key statistics. The following tables present these results, including distributions, document lengths, and vocabulary characteristics.

Table 1 Distribution of Document Types

Type	Count	Percentage (%)
Wikipedia	208	40.62
Blog	110	21.48
GitHub	110	21.48
StackOverflow	84	16.41

This table shows how the dataset is distributed across different document sources. It helps ensure that the dataset includes a variety of content types

Table 2 Distribution of Topics

Topic	Count	Percentage (%)
Artificial Intelligence	93	18.16
Cloud Computing	81	15.82
Cybersecurity	67	13.09
Blockchain	60	11.72
Data Science	59	11.52
Quantum Computing	59	11.52
Internet of Things	47	9.18
Computer Networks	46	8.98

This table presents the number and percentage of documents for each topic, showing how balanced the dataset is across different technical domains

Overall average document length = 1610.23

Table 3 Average Document Length Per Topic

Topic	Average Document Length
Artificial Intelligence	2430.16
Blockchain	1165.28
Cloud Computing	732.12
Computer Networks	2483.78
Cybersecurity	1084.40
Data Science	1716.80
Internet of Things	1693.81
Quantum Computing	1718.73

This table presents the ratio of unique tokens to total tokens for each topic, reflecting the level of vocabulary diversity

Table 4 Token Ratio per Topic

Topic	Total Tokens	Unique Tokens	Token Ratio
Artificial Intelligence	248075	21186	0.0854
Blockchain	74752	7327	0.0980
Cloud Computing	66839	6524	0.0976
Computer Networks	136863	12999	0.0950
Cybersecurity	89198	10250	0.1149
Data Science	124273	13971	0.1124
Internet of Things	86628	8871	0.1024
Quantum Computing	107925	8219	0.0762

This table presents the ratio of unique tokens to total tokens for each topic, reflecting the level of vocabulary diversity

Table 5 Stopword Percentage per Topic

Topic	Total Tokens	Stopword Count	Stopword Percentage (%)
Artificial Intelligence	248075	81384	32.81
Blockchain	74752	28054	37.53
Cloud Computing	66839	21943	32.83
Computer Networks	136863	40403	29.52
Cybersecurity	89198	25627	28.73
Data Science	124273	34946	28.12
Internet of Things	86628	29215	33.72
Quantum Computing	107925	36130	33.48

This table shows the proportion of stopwords in each topic, giving insight into the writing style and level of technical content

Table 6 Lexical Overlap Between Topics

Topic 1	Topic 2	Shared Terms	Total Unique Terms	Lexical Overlap (%)
Artificial Intelligence	Blockchain	4545	23968	18.96
Artificial Intelligence	Cloud Computing	4256	23454	18.15
Artificial Intelligence	Computer Networks	6429	27756	23.16
Artificial Intelligence	Cybersecurity	5457	25979	21.01
Artificial Intelligence	Data Science	7881	27276	28.89
Artificial Intelligence	Internet of Things	4931	25126	19.63
Artificial Intelligence	Quantum Computing	4618	24787	18.63
Blockchain	Cloud Computing	3209	10642	30.15
Blockchain	Computer Networks	3964	16362	24.23
Blockchain	Cybersecurity	3751	13826	27.13
Blockchain	Data Science	3823	17475	21.88
Blockchain	Internet of Things	3396	12802	26.53
Blockchain	Quantum Computing	3093	12453	24.84
Cloud Computing	Computer Networks	4049	15474	26.17
Cloud Computing	Cybersecurity	3633	13141	27.65

Cloud Computing	Data Science	3821	16674	22.92
Cloud Computing	Internet of Things	3409	11986	28.44
Cloud Computing	Quantum Computing	3122	11621	26.87
Computer Networks	Cybersecurity	4888	18361	26.62
Computer Networks	Data Science	5654	21316	26.52
Computer Networks	Internet of Things	4545	17325	26.23
Computer Networks	Quantum Computing	4280	16938	25.27
Cybersecurity	Data Science	4617	19604	23.55
Cybersecurity	Internet of Things	4052	15069	26.89
Cybersecurity	Quantum Computing	3542	14927	23.73
Data Science	Internet of Things	4377	18465	23.7
Data Science	Quantum Computing	4131	18059	22.88
Internet of Things	Quantum Computing	3426	13664	25.07

This table measures the similarity between topics by showing how many terms are shared between them, indicating how closely related the topics are